

Econ560: Problem Set 2

Due: October 28, 11.59pm

The solution must be written in Latex and emailed to rsaggio@mail.ubc.ca.

Question 1: Katz and Murphy (1992)

Consider the constant elasticity of substitution (CES) production function

$$F_t(H_t, L_t) = \left(A_{H_t} H_t^{\frac{\sigma-1}{\sigma}} + A_{L_t} L_t^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad (1)$$

where σ is the elasticity of substitution between high and low skilled labor. For now, assume that labor supply is perfectly elastic.

a) Start by considering some special cases of the CES production function

- Take the limit of F_t as $\sigma \rightarrow 0$. What is the production function?
- Take the limit of F_t as $\sigma \rightarrow \infty$. What is the production function?
- Take the limit of F_t as $\sigma \rightarrow 1$. What is the production function?

b) Assume a perfectly competitive input market. Derive the skill premium $\log \omega_t = \log\left(\frac{w_{H_t}}{w_{L_t}}\right)$ where w_H and w_L are the wages paid to high and low skilled workers, as a function of the exogenous parameter of the model. What is the effect of skill biased technological change on the skill premium, $\frac{\partial \log \omega_t}{\partial \log \frac{A_{H_t}}{A_{L_t}}}$? What is the effect due to an increase in the relative supply of high skilled workers, $\frac{\partial \log \omega_t}{\partial \log \frac{H_t}{L_t}}$?

c) Now suppose that labor supply is elastic. The labor supply functions are given by

$$H_t = H_{0t} w_{H_t}^\epsilon \quad (2)$$

$$L_t = L_{0t} w_{L_t}^\epsilon \quad (3)$$

where ϵ is the elasticity of labor supply. Derive a new expression for the skill premium as a function of the exogenous parameters of the model. What is the effect of skill biased technological change on the skill premium? What is the effect of an exogenous increase in the relative labor supply $\log \frac{H_{0t}}{L_{0t}}$?

d) Suppose that technology evolves according to a stochastic time trend

$$\log \frac{A_{H_t}}{A_{L_t}} = \lambda_0 + \lambda_1 t + e_t \quad (4)$$

where e_t is an iid idiosyncratic error term with variance V_e . The exogenous component of labour supply is given by

$$\log \frac{H_{0t}}{L_{0t}} = \psi_0 + \nu_t \quad (5)$$

where ν_t is another iid idiosyncratic error term with variance V_ν . Consider running a Katz and Murphy style OLS regression of the type

$$\log \omega_t = \beta_0 + \beta_1 t + \beta_2 \log\left(\frac{H_t}{L_t}\right) + u_t \quad (6)$$

Start by assuming that labor supply is elastic. Derive formulas for the population OLS coefficients β_1 and β_2 as a function of the structural parameters of the model. Then consider the cases where (a) $\epsilon = 0$, (b) $V_e = 0$ and (c) $V_\nu = 0$. In what cases is β_2 equal to the inverse of substitution elasticity $-\frac{1}{\sigma}$? In what cases is β_1 equal to the true trend in technology λ_1 ?

e) In view of your results in part (d), comment on the assumptions required for the Katz and Murphy style regression to be meaningful. Do you find these assumptions plausible?

Question 2: Complementarities in Production

Suppose output in a sector is produced according to the following function

$$Q = \left(\sum_i A_i \right)^\alpha \left(\sum_i M_i \right)^\beta \left(\sum_i A_i M_i \right)^{1-\alpha-\beta}$$

where A_i is employee i 's analytical ability and M_i is their motivation. Such a production process exhibits strong complementarities between individual productivities, if one individual has very low ability or motivation it can bring down the output of the entire firm. There are two types of employees: Hippies (H) and Nerds (N). Each Hippy supplies one unit of motivation and two units of analytical ability. Each Nerd supplies one unit of ability and three units of motivation. Note: it is useful to read Rosen (1983, Journal of Human Resources) before attempting this question.

- a) Verify that Q exhibits CRTS in H and N . What is the elasticity of substitution between H and N ?
- b) Assume $\alpha + \beta = 1$. Derive a relationship between the wages of workers and the marginal productivity of skills. Do the same when $\alpha + \beta < 1$. Comment on the usefulness of the notion of skill prices.